Integration Project: Task and Data Brief

Team name / Canvas group: Group 6, Next-Generation Diagnostics

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Goal

Starting task: A simple starting task would be a binary YES / NO tumor classifier. Further exploration would focus on three types of tumor classifications; some members expressed interest in tumor segmentation later on.

Who is the user?	Medical physicians as a diagnostic aid or low-income groups with little access to healthcare as an efficient screening when physicians are unavailable.
Input	MRI images of deidentified patients. They will be subjected to proprocessing and pretrained feature extractors.
Useful output	The simple starting task would output a binary YES / NO for tumor detection. Our later multi-class model would perform multiclass classification using a softmax output layer with four mutually exclusive categories: "Meningioma," "Glioma," "Pituitary," and "No Tumor." If time permits the tumor segmentation model would output a binarized bit "mask" image with a region of 1's outlining where the tumor is.

Example: MRI scan -> Brain tumor on left side

Why this direction: We are interested in the health care space and were leaning in the direction of brain tumors, skin cancer or breast cancer.

Survey

Brain tumor MRI datasets with "Meningioma", "Glioma," and "Pituitary" label classifications were procured from Kaggle and Hugging Face since they offer free, open access. We found a variety of options, but BRISC [1,6] created by Fateh et al, (2026) was very recent and from the reputable journal Scientific Data, so it caught our attention. Its advantages are that it was curated and revised by a radiologist and a physician to ensure correct labeling and prevent data leakage. They also addressed the class imbalance and limited diversity in patient demographics that its predecessors suffered from. BRISC's limitations are derived from its parent dataset BrainTumorMRIDataset. The original dataset was a haphazard collection from multiple public datasets which suffered from severe data leakage between the training and validation set since it had no patient level ID's, and "lacked comprehensive acquisition metadata such as scanner type, field strength, or sequence parameters". After rigorous annotations from experts, the researchers were able to remove most of the duplicated and leaked data, however normalization across MRI hardware and protocols (e.g. only T1-weighted scans were retained in BRISC) was not achieved. As a reference paper, we searched for experiments that used similar data. We found Missaoui et al., (2026) [2] very interesting since it uses a novel approach in the field of brain tumor analysis of incorporating a self-supersiving DINO vision transformer as a pretrained feature extractor, rather than supervised pretrained CNNs like ResNET or EfficientNet. They use Version 2 of BrainTumorMRIDataset, which claims it has addressed the data leakage problems Fateh *et al.*, pointed out. An advantage is the use of techniques we learned in class e.g. ReLU activations, Softmax outputs, and Adam optimizations and the detailed explanations on how to preprocess our data (e.g., Gaussian filtering, morphological operations etc.). A limitation is that Version 2 of the dataset still lacks patient specific IDs, so it has raised concerns that data redundancy or leakage is still possible. Our open-source project is from @abdallahwagih in Kaggle [3], who does the same multiclass model in our goal with a similar dataset using EfficientNet instead of DINO. A limitation is that he uses version 1 of BrainTumorMRIDataset so his accuracy is misleading, but his implementation is thorough and clear to read. However, his sophisticated use of TensorFlow led me to use the TensorFlow tutorials as technical references. [4,5]

Data

Dataset inspected: BRISC: Annotated Dataset for Brain Tumor Segmentation and Classification
https://www.kaggle.com/datasets/briscdataset/brisc2025/data

Example	Input fields (what we saw)	Label / target	Notes
1	Contrast-enhanced T1 MRI;	Tumor present	Original class may be any of

Example	Input fields (what we saw)	Label / target	Notes
	axial plane (looking from above)		glioma, meningioma, or pituitary
2	Contrast-enhanced T1 MRI; coronal plane (looking from front)	Tumor present	Shows another imaging orientation
3	Contrast-enhanced T1 MRI; sagittal plane (looking from side)	Non-tumorous	Uses the BRISC's non-tumorous class

Task fit:

The MRI image serves as the input to the model, while the original tumor_label can be converted to a binary target. Labels like glioma, meningioma, and pituitary become "tumor present", while no tumor becomes "non-tumorous". BRISC is suited for this because of its 6,000 weighted images that are already labeled that span multiple planes for tumor detection.

Uncertainty:

Subject independence is an uncertainty that will have to be acknowledged. The parent dataset our data was derived from did not contain complete patient identifiers, and it isn't guaranteed that all images from the same subject were separated (the physicians who revised it could have missed some). The provided metadata and filenames will need to be inspected before relying on the supplied split.

References

[1] "BRISC: Annotated Dataset for Brain Tumor Segmentation and Classification" Fateh et al., 2026

[2] Missaoui, R., Del Coco, M., Saadaoui, W., Hechkel, W., Helali, A., Carcagnì, P., & Leo, M. (2026). Brain Tumor Classification Using DINO Features and Lightweight Classifiers. *Electronics*, *15*(5), 952. https://doi.org/10.3390/electronics15050952

[3] Ibrahim, A. W. (2023). Brain tumor detection [Kaggle notebook]. Kaggle. https://www.kaggle.com/code/abdallahwagih/brain-tumor-detection

[4] TensorFlow. TensorFlow documentation. https://www.tensorflow.org/tutorials

[5] darkhelmet. TensorFlow- Cheat Sheets. Cheat Sheets. https://darkhelmet.github.io/cheats/python/tensorflow/

[6] Amirreza Fateh, Yasin Rezvani, Sara Moayedi, Sadjad Rezvani, Fatemeh Fateh, Mansoor Fateh, and Vahid Abolghasemi. (2025). BRISC 2025 [Dataset]. Kaggle. https://doi.org/10.34740/KAGGLE/DS/7632487